TEACH YOURSELF INTERMEDIATE XII UNIT 5

Applications & Impacts (AI & Security)

Neural networks, collaboration, and threats

What you will master

- Outline neural networks and deep learning uses
- Explain secure collaboration and data protection
- Identify threats (phishing, malware) and mitigations
- Discuss equity and access in digital tools

How to use these notes

- Read one section, then close the book and explain it aloud.
- Do every *Try it* box before moving on.
- Memorize tables and truth patterns — exams love them.

5.1 Neural networks (conceptual)

Neurons receive inputs, apply weights, activate, and pass signals layer-by-layer. **Deep learning** uses many layers for vision, speech, and language tasks.

Needs large datasets, compute power, and ethical oversight (bias, privacy).

5.2 Security & collaboration

Use strong passwords, 2FA, encryption in transit (HTTPS), least-privilege sharing. Threats: phishing, ransomware, social engineering.

Mitigate: updates, backups, user training, firewalls, antivirus.

5.3 Digital equity

Equal access means affordable connectivity, accessible interfaces, and policies that don’t exclude rural or disabled learners.

Teach Yourself — Practice & Check

1. List three deep-learning apps and the data they need.
2. Design a phishing-awareness poster for classmates.